BASICS OF STATISTICS

TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 1

5-MINUTE REVISION

Source: Chapter 1 of the textbook.

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1. DEFINITION

Statistics is the science which deals with the

1. Collection of data
2. Organization of data or classification of data
3. Presentation of data
4. Analysis of data
5. Interpretation of data

2. SCOPE AND IMPORTANCE

Functions of statistics: Statistics plays a crucial role in simplifying complex data, transforming it into clear and meaningful information.

It supports decision-making by presenting facts in an organized manner, aids in the formulation of effective policies, facilitates comparisons, and assists in making forecasts.

By applying appropriate statistical methods, researchers can draw valid conclusions from experiments.

3. BRANCHES OF STATISTICS

Statistics

- Descriptive statistics, which deals with summarizing data from a sample or a population using indexes or measures such as the mean or standard deviation etc.
- Inferential statistics, use a random sample of data taken from a population to describe and make inferences about the population parameters.

4. POPULATION AND SAMPLE

Population
A population is the complete group of interest

Sample
a sample is a part selected for study.

A parameter describes a population, while a statistic is calculated from sample observations.

5. TYPES OF VARIABLES

Type	From the textbook
Quantitative	A quantitative variable is one that can be expressed in numerical terms and takes values that are measurable.
Discrete	Discrete variables are variables that can only take a finite or countable number of distinct values.
Continuous	Continuous variables, on the other hand, are variables that can take any value within a given range or interval and can be measured.
Categorical	Categorical variables classify observations into groups.

6. COLLECTION OF DATA

Primary data
Primary data are first-hand, original data collected directly from a single source by the researcher for a specific purpose.

Secondary data
In contrast, secondary data are obtained from existing sources or agencies and have been previously collected and processed for different purposes.

7. MEASUREMENT SCALES

Nominal	Data measured on the nominal scale consist of categories that are distinct but have no inherent order or ranking.
Ordinal	This means that the categories have a specific rank or order, but the differences between the categories are not necessarily uniform or meaningful.
Interval	On the interval scale, the data points are ordered, and the differences between them are meaningful and measurable.
Ratio	The ratio scale is the highest level of measurement and applies to quantitative data.

8. CONSTRUCTING A GROUPED DISTRIBUTION

1. Decide the number of classes you wish to construct
2. Determine the class width
3. Set up the individual class limits
4. Tally the items into the classes
5. Count the number of items in each class

9. WORKED TABLE FROM THE BOOK

an agricultural student measured the lengths of leaves on an oak tree (to the nearest cm).

Class	Frequency
1-3	3
4-6	6
7-9	10
10-12	5
13-15	5
16-18	9

Source: Chapter 1, frequency distribution table.

10. IMPORTANT FORMULAE

Number of classes using Sturges' formula

$$k = 1 + 3.322 \log_{10} N$$

Approximate class width

$$C = \frac{|\max - \min|}{k}$$

Class mark

$$\text{Class mark} = \frac{\text{Lower class limit} + \text{Upper class limit}}{2}$$

Class width

$$C = \text{Upper class boundary} - \text{Lower class boundary}$$

Relative frequency

$$\text{Relative frequency} = \frac{f}{N}$$

Sum of frequencies

$$\sum f = N$$

11. KEY POINTS AND LIMITS

Common limitations of statistics: Statistical conclusions depend on the quality of the data, the design of the study and the assumptions of the method.

A large dataset cannot by itself correct biased selection or poor measurement.

Sample results are subject to uncertainty, and an association alone does not establish cause and effect.